

A Viseme-Based Approach to Emotion-Consistent Facial Expression Generation in Humanoid Head Robots: Synchronizing Lip Shape, Speech, and Emotional Expression

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Abstract—This paper presents a viseme-based approach for generating coordinated facial expressions and lip movements in humanoid robots. The method integrates facial expression control with synchronized lip movements to address the issue of dynamic lip movements affecting emotional expression during speech. It ensures precise synchronization between lip movements and speech, while incorporating dynamic changes in other facial features, such as eyebrows and eyes, thereby enhancing the accuracy of emotional expression. The approach uses eight basic visemes to align lip movements with speech across ten emotional states (neutral, serious, happy, bored, angry, disgusted, fearful, sad, tense, and surprised). Additionally, psychological theories are applied to refine the control of eyebrow and eye movements, improving the naturalness and emotional accuracy of the robot's expressions. Cognitive evaluation experiments demonstrate that the overall accuracy of facial expression is 94%, while the synchronization accuracy rate between emotional lip movements and speech is 74%. In comparison to speech-driven lip synchronization, this method exhibits higher emotional expression consistency and a greater variety of expressions. Future research should focus on improving dynamic emotion expression and synchronization in speech and facial movements for more natural human-robot interaction.

Keywords—Humanoid Head Robots, Viseme, Facial Expression Generation, Speech and Facial Expression Synchronization, Emotional Expression, Lip Movement Synchronization

I. INTRODUCTION

With the rapid development of artificial intelligence and robotics technologies, humanoid emotion-interactive robots are gradually being applied across various fields. Despite significant progress, a key challenge that remains unresolved is how to present natural, coherent, and emotionally rich dynamic expressions during voice interaction. In particular, the coordination of lip movements with other facial expressions has become a critical issue in humanoid robot facial expression generation[1].

In emotional interactions, lip movements are a key element of emotional expression and are closely linked to speech production. Simply adjusting lip movements may cause misalignment with other facial features, disrupting expression consistency and emotional accuracy. Current research on humanoid robot facial expressions mainly focuses on static expressions, leaving the coordination of lip movements with facial expressions largely unresolved. Specifically, synchronizing lip movements with emotional continuity in speech contexts remains a significant technical challenge[2]. While some studies have attempted to decouple action units to address this issue[3], the relationship between lip movements and speech is often underexplored, leading to inconsistencies between emotional expression and lip synchronization, thus affecting emotional accuracy. Additionally, while deep neural networks have been used to generate lip-syncing facial animations for emotional speech[4][5], these methods still face limitations in expressiveness and synchronization, and their facial animations struggle to align with actual robot control systems, restricting their application in humanoid robots. Therefore, current methods for generating emotional expressions still need improvement in integrating lip movements with other facial features (such as eyebrows, eyes, and cheeks), especially in dynamic scenarios that require speech and emotional synchronization.

This paper proposes a novel method for generating coordinated facial expressions, lip movements, and emotional expressions. The method ensures precise lip-speech synchronization and coordinates facial features (such as eyebrows and eyes) with emotions, maintaining emotional consistency during speech. The approach uses eight visemes for lip movement control, aligning lip movements with speech. Lip control parameters are adjusted for ten emotional states (neutral, serious, happy, bored, angry, disgusted, fearful, sad, tense, and surprised), ensuring facial feature coordination to enhance naturalness and emotional accuracy. Additionally, this research

applies psychological theories to design control rules for the eyebrows and eyes, refining the robot's facial expression generation. These parameters account for emotion type and intensity, ensuring coherent and vivid emotional expression.

II. VISEME-BASED HUMANOID HEAD ROBOT EMOTIONAL SPEECH LIP SYNCHRONIZATION

A. Overview

The method proposed in this paper consists of two primary stages: the establishment of the emotional viseme library and the generation of facial expressions synchronized with emotional speech and lip movements, as illustrated in Fig. 1.

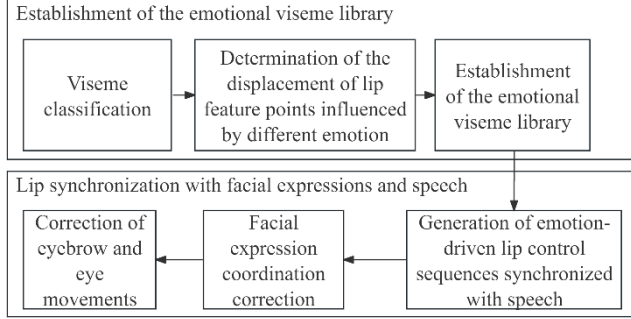


Fig. 1. Viseme-based Humanoid Head Robot Emotional Speech Lip Synchronization Method

B. Determination of Basic Visemes

Visemes represent the visual configurations of articulatory organs (e.g., lips, tongue, jaw) corresponding to phonemes, capturing the dynamic continuity of facial movements. This study modifies the existing 11-viseme model for Mandarin Chinese[6], adapting it to meet the engineering constraints of robotic lip control and the requirements for emotional expression. Based on the robot's facial structure and dynamic lip movement correction for emotional interactions, an 8-viseme classification is proposed, ensuring precise mechanical movement and enabling parameterized control for emotional lip representations like smiling and surprise. The eight basic visemes are presented in Table I.

C. Establishment of the Emotional Viseme Library for Lip Movements

In human-robot interaction, the robot responds to human language, actions, or commands, such as answering questions. The response is first broken down into phonetic units, which are input into the humanoid head robot's lip movement system. The system constructs a time sequence for each phoneme based on its pronunciation, and the phonemes are played sequentially through a sound output device to produce speech. To improve viseme representation in the robot, a quantitative description of the visemes is required, followed by the creation of a basic viseme library[6]. This is achieved by recording a person reading phonemes corresponding to different visemes across ten emotional states. Using the eight Chinese visemes defined in Table I, typical images are extracted from the recordings. Image processing identifies feature points for each emotion, and based on their coordinates and corresponding visemes, a basic viseme library is built. This library serves as a reference for generating

TABLE I. EIGHT BASIC VISEMES AND THEIR CORRESPONDING PHONEMES

No.	Visemes	Phonemes
0		p,b,m,basic
1		f
2		t,d,n,l,ch,r,sh,zh
3		w
4		k,g,h,eng,e,en,ing
5		j,q,x,y,i,z,c,s
6		o,ong
7		an,a,ang

facial expressions in the robot. The feature points' positions for each viseme vary across emotional states, and the basic viseme library is represented as:

$$V_f = \{V_k^f\} \quad (k = 0, 1, 2, \dots, 7; f \in \{f_1, f_2, \dots, f_m\}) \quad (1)$$

In this equation, f represents the set of feature point coordinates, m denotes the number of feature points, and V_k^f indicates the set of feature point coordinates corresponding to the k -th viseme. In the emotional viseme library, the coordinates of the feature points change with the emotional state E . Each emotional state corresponds to a distinct facial expression, and this variation in facial expression directly affects the viseme features of each phoneme. Specifically, the impact of the j -th emotional state on the feature point coordinates can be represented by a correction term $\Delta f_i(E_j)$, which reflects the change in the coordinates of the feature point f_{ji} relative to the coordinates f_{ji}^{base} in the baseline emotional state. Based on this, the coordinates of feature point f_{ji} in emotional state E_j can be expressed as:

$$f_{ji} = f_{ji}^{base} + \Delta f_i(E_j) \quad (i = 0, 1, 2, \dots, m; j = 0, 1, 2, \dots, 9) \quad (2)$$

Based on (1) and (2), the emotional viseme library is constructed, which includes the feature points of each viseme for all emotional states. The emotional viseme library is as follows:

$$V_{fE} = \{V_{jk}^{f_e}\} \quad (i = 0, 1, \dots, 7; j = 0, 1, \dots, 9; f_e \in \{f_{j1}, f_{j2}, \dots, f_{jm}\}) \quad (3)$$

In this equation, f_e represents the set of feature point coordinates corresponding to all emotional states, m denotes the number of feature points, and $V_{jk}^{f_e}$ indicates the set of feature point coordinates for the k -th viseme under a specific emotional state j . This library contains a total of 80 emotional visemes, corresponding to 8 basic visemes for each of the 10 emotional states.

Analysis of synergistic phonetic effects in daily Mandarin conversation shows that viseme generation follows specific dynamic assimilation patterns. When viseme differences occur

within a syllable, two typical articulatory movement patterns emerge. The first is reverse assimilation, where a smaller viseme at the syllable's end overrides the larger one at the beginning, as seen in the /du/ combination, where the final /u/ dominates despite the presence of /d/. The second is forward transition, where a larger final phoneme leads to a rapid transition from a smaller to larger viseme, as in the /wa/ combination, where the lips significantly move outward from /w/ to /a/. Thus, the determination of visemes in multi-phoneme combinations can be simplified to the rule of extreme viseme values at the start and end: for any syllable $C_1C_2...C_n$, its core visual feature is determined by the difference between the first phoneme $\phi(C_1)$ and the last $\phi(C_n)$. If $\phi(C_n) > \phi(C_1)$, a linear expansion occurs; otherwise, the final viseme covers the initial one. Each syllable requires a maximum of two visemes for the phoneme-to-viseme transformation. During speech playback, synchronized lip movements, aligned with phoneme timestamps and the viseme-to-phoneme mapping in Table I, generate continuous viseme representations.

D. Implementation of the Basic Emotional Viseme Model for Humanoid Head Robots

When defining the robotic physical lip shapes corresponding to the basic visemes in Table I, the first step is to determine the coordinates of the feature points for each basic viseme and their motion trajectories. Based on this, we consider using a scaling factor k_p to coordinate the size ratio between the robot and the subject being measured (typically a human), ensuring that the robot's lip shape in a natural state closely matches the shape of the human lip shape. Specifically, the scaling factor k_p is defined as:

$$k_p = \frac{L}{L_l} \quad (4)$$

In this equation, L represents the distance between the corners of the robot's mouth when in a natural state with the lips closed, while L_l denotes the width between the corners of the subject's mouth (typically a human) in a natural state. By setting this scaling factor, an effective size match between the robot's lip shape and the human lip shape can be achieved. Ultimately, the robot's emotional viseme model can be expressed as:

$$R_{V_{jk}}^{d_{oe}} = k_p V_{jk}^{f_e} \quad (5)$$

In this equation, $R_{V_{jk}}^{d_{oe}}$ represents the emotional viseme model for the robot when the control points are located on the lips, and $d_{oe} \in \{d_{oj1}, d_{oj2}, \dots, d_{ojn}\}$ corresponds to the robot's emotional lip movement control points. Here, n is the total number of control points, matching the number of feature points. The control of the robot's lip movements depends not only on lip motion but also on coordination with other facial regions, such as the apple muscles, dimples, and chin. By relocating the control points to these areas, the coordinated movement of facial muscles can be more accurately simulated, enhancing the integration of facial expressions and lip movements within the existing mechanical structure. Specifically, the upper lip control point is moved beneath the apple muscles, the mouth

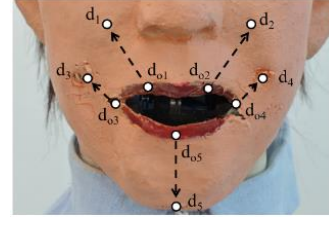


Fig. 2. Migration of Lip Control Points in Humanoid Robots

TABLE II. BASIC LIP MOVEMENT MODEL FOR THE HUMANOID HEAD ROBOT UNDER A NEUTRAL EMOTIONAL STATE

Visemes' No.	d_1	d_2	d_3	d_4	d_5
0	0	0	0	0	0
1	2	2	0	0	1
2	4	4	0	0	3
3	-5	-5	3	3	0
4	0	0	6	6	7
5	-2	-2	8	8	5
6	5	5	-5	-5	9
7	3	3	-2	-2	9

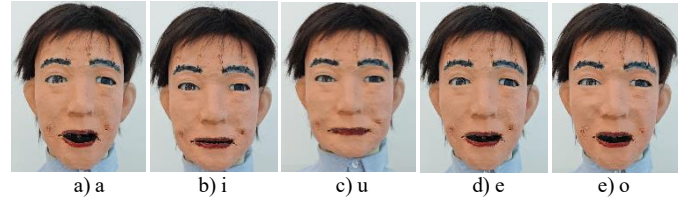


Fig. 3. Photograph of the Basic Lip Movement Reproduction Experiment for the Humanoid Head Robot under a Neutral Emotional State

corner control point is relocated to the dimple area, and the lower lip control point is moved to the chin, as shown in Fig.2.

By adjusting the relative positions and dynamic weights, a displacement mapping relationship is established between the original control points $d_{oe} \in \{d_{oj1}, d_{oj2}, \dots, d_{ojn}\}$ and the control points after migration $d_e \in \{d_{j1}, d_{j2}, \dots, d_{jn}\}$, resulting in the emotional viseme model after the control point migration.

By calculating the average from a large set of measurement data, the emotional viseme model for the humanoid head robot was developed. Table II presents the basic lip movement model data for the humanoid head robot under a neutral emotional state. Fig. 3 shows a photograph of the basic lip movement reproduction experiment under the neutral emotion, based on the lip model in Table II.

III. FACIAL EXPRESSION GENERATION FOR COORDINATING EMOTIONAL LIP MOVEMENTS IN HUMANOID HEAD ROBOTS

A. Theoretical Framework for Facial Expression Generation in Humanoid Head Robots

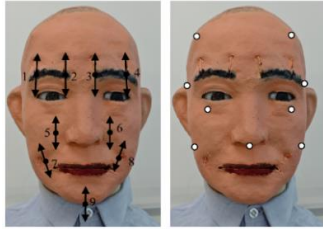
The research of human facial expressions dates back to the 1960s, when Swedish anatomist Hjortsjö first introduced and studied the smallest units of human facial expressions. Building on this research, Ekman and Friesen further investigated the movements of facial muscle groups and their role in expressions, leading to the development of the Facial Action Coding System (FACS)[7], a system designed to describe the

TABLE III. RELATION OF AU AND FACIAL EXPRESSION

AU's No.	Facial Expression Change	AU's No.	Facial Expression Change
1	Inner brow raiser	12	Outer lip corners pull up
2	Outer brow raiser	15	Lip corners pull down
4	Brow lower	23	Lips tighten
5	Upper eyelid raise	24	Lips press together
6	Cheeks raised	25	Lips parted
7	Eyelids tight	26	Jaw lower
9	Nose bridge wrinkling	43	Lower eyelid drooping

TABLE IV. AU COMBINATIONS AND CONTROL POINTS FOR CORRESPONDING BASIC FACIAL EXPRESSIONS

Expression	AU Combinations	Control Points
Serious	4+7+24	1,2,3,4,5,6,7,8,
Happy	6+12	5,6,7,8,
Bored	4+15+43	1,2,3,4,5,6,7,8,
Angry	2+5+7+23	1,2,3,4,5,6,7,8,9
Disgusted	9+15	5,6,7,8
Fearful	1+4+5+25	1,2,3,4,5,6,7,8,9
Sad	1+4+15	1,2,3,4,7,8,
Tense	4+5+23	1,2,3,4,5,6,7,8,
Surprised	1+2+5+26	1,2,3,4,5,6,9



a) Control points b) Fixed points

Fig. 4. Facial control points and fixed points in humanoid head robots

variations in facial expressions. FACS divides the human face into 44 independent action units (AUs), and analyzes the role of these action units in controlling facial expressions and their corresponding facial regions, as shown in Table III.

The humanoid head robot "H&Frobot-III" used in this research adopts Ekman's FACS theory, selecting 14 action units (AU1, AU2, AU4, AU5, AU6, AU7, AU9, AU12, AU15, AU23, AU24, AU25, AU26, AU43) as facial expression motion units based on the robot's specific needs. These AUs correspond to the robot's facial expression control points, with their positions and motion directions shown in Fig. 4a) (indicate positions, and arrows show motion directions). The fixed points, shown in Fig. 4b), represent locations where the facial elastic skin is anchored to the facial shell, unaffected by control points. By adjusting the combination and displacement of these control points, the robot can generate various basic facial expressions. The AU combinations for basic expressions and their corresponding control points are listed in Table IV.

B. Implementation of Coordinated Emotional Lip Movement and Facial Expression in Humanoid Head Robots

In emotional lip movement design, lip shape changes must be coordinated with the movement of the eyebrows, eyes, and cheeks, as these facial features significantly affect lip movements. Specifically, the motion of these features not only influences the lip expression directly but also creates a

synergistic effect, especially in dynamic emotional expressions. Therefore, to ensure natural and accurate expressions, it is essential to precisely calculate the eyebrow and eye control parameters for each emotion by integrating lip control points and fixed points. In this research, the determination of eyebrow and eye control parameters is based on the pre-planned emotional lip movement design. Control point movements are driven by calculating the force required for point displacement to achieve the desired facial expression changes.

Based on Table IV, the relationships between emotional lip movements and eyebrow control points were adjusted for the 10 emotional states defined in this study. By analyzing and fine-tuning the stretching control points, the appropriate eyebrow and eye control values were optimized to align with dynamic lip movements. This process improves both the naturalness of the robot's expressions and the accuracy of emotional expression. The method enhances expression control precision while ensuring coordination across facial features during the robot's complex emotional expressions.

C. Facial Expression Control Strategy for Coordinating Lip Movements in Humanoid Head Robots

After generating the emotional control sequences corresponding to speech, further adjustments were made to the facial expression control sequences for coordinating emotional lip movements, in order to enhance the naturalness of speech expression. These adjustments were based on psychological theories, with particular focus on the dynamic features of the eyebrows and eyes across different emotional states. The movement characteristics of the eyebrows primarily manifest in their amplitude and frequency, with significant differences in eyebrow movement patterns across emotional states[8]. For example, during emotions such as anger and surprise, eyebrow movements are frequent and have a large amplitude, often accompanied by rapid upward or downward shifts. In contrast, during sadness, eyebrow movements have a smaller amplitude, a lower frequency, and typically display a smooth, slow downward motion. The dynamic features of the eyes work in close coordination with the eyebrows, involving changes in gaze duration, movement amplitude, and eye blink frequency[9]. In anger, the frequency of eye movement is higher, with rapid shifts, whereas in positive emotions, the eye gaze is longer, with larger movement amplitude, exhibiting more relaxed characteristics. During states of tension or anxiety, frequent eye revisits and short gaze durations reflect the instability of the emotional state. Table V illustrates the eyebrow motion features for the 10 basic emotional states, while Table VI presents the corresponding eye motion features.

TABLE V. EYEBROW MOTION FEATURES

Emotion	Movement Amplitude	Movement Frequency
Neutral	Small	Low
Serious	Small	Low
Happy	Medium	Medium
Bored	Large	Low
Angry	Large	High
Disgust	Medium	High
Fearful	Medium	Medium
Sad	Small	Low
Tense	Medium	Medium
Surprised	Large	High

TABLE VI. EYE MOTION FEATURES

Emotion	Offset Direction	Gaze Duration	Movement Amplitude	Movement Frequency
Neutral	Random	Medium	Small	Low
Serious	None	Long	None	None
Happy	Diagonal Upward	Long	Medium	Low
Bored	Diagonal Downward	Short	Small	Low
Angry	Up	Long	Large	Low
Disgust	Left	Short	Small	High
Fearful	Down	Medium	Large	Medium
Sad	Down	Medium	Small	Low
Tense	Left-Right	Short	Medium	Medium
Surprised	None	Long	Large	Low

Based on the motion features in Tables V and VI, the target emotional control sequence was adjusted to enhance the coordination between the humanoid head robot's facial expressions and emotional lip movements.

IV. EXPERIMENTAL RESEARCH ON COORDINATING LIP MOVEMENTS WITH FACIAL EXPRESSIONS IN HUMANOID HEAD ROBOTS

A. Development and Improvement of Humanoid Head Expression Robots

The humanoid head robot H&Frobot-III[10] consists of four main components: the head organ motion mechanism, facial expression drive mechanism, facial shell, and elastic facial skin. The head organ motion mechanism includes three parts—eyelid, eyeball, and jaw drives—powered by servo motors to simulate human eye, eyelid, and jaw movements. The facial expression drive mechanism includes drives for the inner and outer eyebrows, cheeks, and mouth corners, simulating human eyebrow and cheek movements. Additionally, the robot is equipped with two CMOS micro-cameras in its eyes for binocular vision, capturing facial expression data from the interaction subject. The head also integrates a microphone and speaker to simulate auditory and speech expression capabilities, enhancing its bionic effect for more realistic sensory interactions.

B. Experimental Research on Coordinated Lip Movement Expression

This experiment aims to verify whether the facial expressions and lip movements of the robot can be well coordinated under 10 different emotional states, ensuring the natural reproduction of coordinated speech lip movements. Based on interaction instructions, the robot will verbally articulate statements corresponding to each of the 10 emotional states: neutral, serious, happy, bored, angry, disgusted, fearful, sad, tense, and surprised. It will then express the corresponding facial expressions and lip movements for each emotional state. Partial experimental results are shown in Fig.5, demonstrating the coordinated speech lip movement expression of the humanoid head robot H&Frobot-III across various emotional states. After cognitive evaluation of the video clips by 10 participants, the expression accuracy and lip synchronization for each emotional state were assessed. The evaluation results are presented in Table VII.

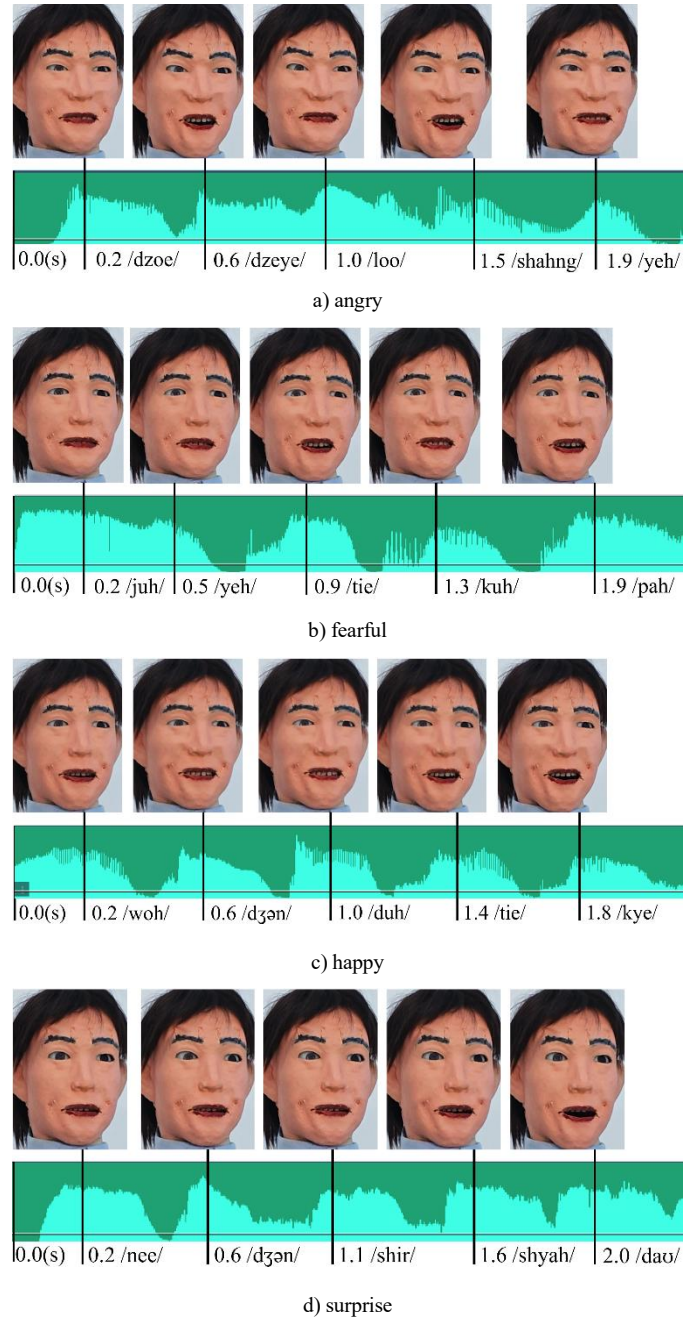


Fig. 5. Coordinated speech lip movement expression of the humanoid head robot H&Frobot-III across various emotional states

TABLE VII. COGNITIVE EVALUATION RESULTS OF COORDINATED SPEECH LIP MOVEMENT EXPRESSION UNDER 10 EMOTIONAL STATES

Emotion	Expression Accuracy	Lip Synchronization
Neutral	100%	60%
Serious	80%	80%
Happy	100%	100%
Bored	100%	80%
Angry	100%	80%
Disgust	80%	40%
Fearful	80%	100%
Sad	100%	60%
Tense	100%	100%
Surprised	100%	40%

V. DISCUSSION

This research introduces a method for achieving emotion-driven lip synchronization in humanoid robots through a custom emotion-viseme database. The approach ensures effective coordination between facial expressions and lip movements, enhancing emotional consistency during human-robot interactions (HRI).

The cognitive evaluation results indicate that emotions such as Neutral, Happy, Bored, Angry, Sad, Tense, and Surprised exhibit 100% facial expression accuracy, while Serious, Disgust, and Fearful show lower accuracy (80%), likely due to the subtlety of their facial features. Lip synchronization is highest for Happy, Fearful, and Tense (100%), whereas Sad, Disgust, and Surprised exhibit lower synchronization rates (60%, 40%, and 40%, respectively), which may be attributed to less expressive facial movements. Similar research on speech-driven coordination of facial expressions and lip movements in animatronic robots primarily focuses on the natural synchronization between facial expressions and lip movements[11]. However, their research does not include cognitive evaluation of the accuracy of specific emotional expressions. In contrast, our research demonstrates an overall facial expression recognition accuracy of 94%, with a synchronization rate of 74% between emotional lip movements and speech. The emotion-driven viseme-based approach proposed in this research enables the robot to express specific emotional states, thereby providing a more accurate and targeted solution for emotional interaction.

Compared to other speech-driven studies, most research on facial expression animation with lip synchronization focuses on CG-based avatars[12], which perform well in controlled environments but face challenges when applied to real robots due to limitations in system precision. In contrast, our approach is specifically designed for actual humanoid robot structures, making it more suitable for real-world applications. In addition, while other researches on robotic facial expression imitation concentrate primarily on replicating human facial dynamics[13][14][15], this research integrates speech-driven emotion synchronization, ensuring that both facial expressions and lip movements are aligned with emotional content, offering a more comprehensive solution for emotional interaction in robots.

This work has practical applications in fields like social robotics, where it could enhance user engagement in elder care, education, and mental health support. Additionally, it could improve service and entertainment robots, creating more immersive and emotionally intelligent interactions.

VI. CONCLUSIONS

The synchronization of emotional lip movements and facial expressions with speech in humanoid head robots is an essential component of natural human-robot interaction. Based on this research, significant progress has been made in coordinating emotional lip movements and facial expressions in humanoid robots. By precisely controlling the alignment between lip movements and speech, and incorporating dynamic changes in facial features such as eyebrows and eyes, the proposed method significantly enhances the naturalness and consistency of the

robot's emotional expression. Experimental results show that while most emotional expressions exhibit high expression accuracy, there is still room for improvement in lip-synchronization across some emotional states. Future research should further optimize the coordination between lip movements and facial expressions in dynamic emotional expressions, especially in terms of speech and facial motion synchronization, to achieve more natural and realistic human-robot interaction.

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